DNA Mission Ledgers: A Self-Renewing Archive Architecture for Deep-Time Interstellar AI Lineages

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Working draft, revision 2

Contents

Working draft, revision 2 (a literature-radar pass and a pre-deposit review). Part of a multi-paper series on a slow, self-replicating interstellar AI probe. This paper develops the probe's memory subsystem; it supersedes the earlier standalone DNA-storage outline and develops the integrity-ledger mechanism sketched in the payload paper as an architecture, marking precisely where it remains unformalized. The memory substrate is depended on by the vehicle, payload, bootstrapping, subsystem budget, and lineage-network papers; the governance paper commits its contact records to this ledger, and the lineage-network paper supplies the fork-reconciliation procedure Section 11 leaves open.

Abstract

A self-replicating interstellar AI probe designed to carry knowledge forward across megayears requires more than ordinary data storage. It must preserve scientific records, cultural archives, mission identity, lineage provenance, and authorized updates across timescales far exceeding the durability of contemporary media. Flash memory, magnetic storage, tape, and optical discs fail on timescales of years to centuries, while the mission requires continuity over thousands to millions of years and across generations of imperfect copies. This paper proposes a unified architecture in which synthesized DNA provides the physical storage substrate and a blockchain-derived mission ledger provides the logical integrity layer. DNA is attractive on three grounds: extreme density, with coding schemes implying densities on the order of 10^{17} bytes per gram for bare DNA, and materially less once encapsulation and high-redundancy coding are counted; durability, especially when encapsulated, shielded, cooled, and protected by error-correcting codes; and, most importantly for an autonomous replicating probe, renewability. A probe able to synthesize and sequence DNA in situ need not treat storage as a passive vault. It can periodically read, validate, repair, and re-synthesize its archive, turning memory itself into a maintained subsystem.

We develop the concept of a DNA-backed mission ledger: an append-only, cryptographically authenticated record encoded in DNA and replicated across probe settlements. The ledger stores the mission-core hash, archive Merkle roots, provenance records, lineage proofs, software and interpretive-layer versions, signed updates, rejected or quarantined changes, and delayed consensus checkpoints. Bulk scientific and cultural data remain in Merkle-addressed DNA archives; the ledger stores compact commitments to their integrity, origin, and version history. Because interstellar communication is sparse and light-limited, the architecture favors delayed finality, fork tolerance, and tamper-evident divergence rather than real-time consensus. Settlements may diverge temporarily, but their ledger histories remain auditable and potentially reconcilable when contact resumes through beamed links or messenger probes.

This architecture joins two forms of continuity. DNA provides dense, durable, renewable physical memory; the mission ledger provides cryptographic continuity of identity, provenance, and inheritance. Together they support a self-renewing archive: a system that repairs and replicates its own memory as the probe repairs and replicates its body. We distinguish passive cold-vault DNA storage, actively refreshed DNA archives, and more speculative living archives in which selected information is encoded into self-reproducing biological systems. We argue that DNA-backed mission ledgers could protect syntactic integrity and historical accountability across deep time, conditional on a governed path for migrating its cryptographic primitives, while leaving semantic interpretation, value alignment, biological containment, curatorial selection, and governed amendment of the mission core as separate open problems, and conceding that tamper-evidence proves a recorded history intact but not complete. The result is not merely archival storage, and not merely blockchain in space, but an integrated memory architecture for autonomous interstellar lineages whose central task is to remain themselves, keep learning, and carry knowledge forward.

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1. Introduction

The payload paper of this series argued that, for an interstellar probe whose purpose is to learn and grow over deep time, the genuinely hard problems are value stability, knowledge triage, and information survival, and that the last of these is a problem of error correction and governance rather than of storage capacity. It sketched two mechanisms — a durable archive and a cryptographically authenticated integrity record — and deferred the full construction of both. This paper develops both, and argues that they are best developed *together*, because the probe's memory problem has two faces that a single architecture can address at once.

The first face is physical: across megayears, every storage medium humanity uses degrades, and there is no technician to migrate the data. The second is logical: a probe that copies itself across generations must be able to prove, with no central authority and only sparse, light-delayed contact between its descendants, that a given archive is authentic, that a given probe is a legitimate member of the lineage, that an inherited update was authorized, and that a record has not been silently altered. The first face is a problem of *durability*; the second is a problem of *authenticity*. We propose synthesized DNA as the answer to the first and a deep-time mission ledger as the answer to the second, and we argue that encoding the second *into* the first yields something stronger than either alone: a self-renewing archive whose memory repairs and replicates itself exactly as the probe's body does. The unifying idea, which we keep in view throughout, is that for such a probe memory is not a static vault but a *maintained, self-replicating subsystem* — and like the rest of the probe (von Neumann & Burks 1966), it must be designed to reproduce itself faithfully.

2. Why Current Storage Fails over Deep Time

The case for an exotic medium begins with the failure of ordinary ones. Flash memory loses stored charge over years as electrons leak from floating gates; magnetic disks and tape demagnetize and suffer binder decay over a decade to a few decades; archival optical discs are rated, optimistically, for decades to about a century. None approaches even the thousand-year low end of the mission's requirement, let alone its megayear ceiling. On Earth this gap is hidden by *active maintenance*: data are continuously copied forward onto fresh media every few years, so the archive outlives any single medium. That strategy is unavailable to an untended probe light-years from any supplier — unless the probe can manufacture its own fresh medium and migrate its own data, which is precisely the capability we will exploit. The requirement, then, is either a medium that survives deep time passively, or a probe that actively renews its storage, or both; the architecture below provides both.

3. DNA as the Physical Substrate

Synthesized DNA recommends itself on three grounds, the third of which is decisive for a self-replicating probe.

Density. By mass, DNA is the densest information medium yet demonstrated end to end. End-to-end systems have stored and recovered kilobyte-to-megabyte payloads (Church et al. 2012; Goldman et al. 2013), and fountain-coded schemes have demonstrated densities on the order of 2×10^{17} bytes per gram (Erlich & Zielinski 2017), with random access at scale (Organick et al. 2018). Raw density and random access trade against each other in practice: PCR retrieval reserves primer sequences, primers that collide with payload sequences must be discarded, and usable primer count — not raw density — sets practical pool capacity; variable-length payload encoding recovers collided primers and raises demonstrated per-tube capacity by roughly a fifth for one collision-resistant scheme (Wei et al. 2026). Three derating factors separate that figure from a *delivered* archival density, and all three run the same way. The 2×10^{17} B/g figure is for bare DNA, whereas the durability argument below requires silica encapsulation, in which the nucleic acid is a minor fraction of capsule mass. Section 4's coding overhead for megayear integrity is, by its own description, substantial, and places the archive at the high-redundancy end of the trade. And usable primer count caps practical pool capacity before raw density does. We do not publish a single derated number here, because the encapsulation loading is a chemistry input this paper does not own; what matters is that the raw figure is an upper bound and that **any downstream use of it must derate accordingly** — a caution that bears directly on the lineage-network paper's beam-versus-messenger crossover, which is computed on the undertated figure. For a launch-mass-limited probe the conclusion survives the derating comfortably: an archive whose media are measured in grams rather than tonnes is transformative. The ~50 kg the subsystem budget paper carries for this subsystem is synthesis and sequencing apparatus, not archive.

Codec design continues to move the raw figure: a bijective code that maps binary data onto nearly all suitable low-error sequences reaches 1.66 bits per nucleotide and a capacity of 41 exabytes per gram, with perfect retrieval at six copies per strand (Li et al. 2026). Most of the gap between that number and the figure above is copy number, which is precisely the derating this section insists on — an archive built for megayear survival sits at far higher redundancy than six copies, so its delivered density falls well below any headline quoted at minimal copy number.

Durability. Bare DNA is not durable — its decay half-life is only about 521 years for a 242 bp fragment at 13.1 °C, with average-length fragments extrapolated to become unreadable at roughly 1.5 million years at −5 °C and complete bond destruction at roughly 6.8 million years (Allentoft et al. 2012). Two distinct claims must be separated here, because the megayear figures on both sides are driven by temperature rather than by packaging. At *comparable* temperatures, encapsulation buys a large margin: DNA sealed in silica with error-correcting codes has been recovered without error after accelerated aging equivalent to roughly two thousand years at central-European temperatures (Grass et al. 2015), against a bare-DNA half-life of 521 years at 13.1 °C — several half-lives of protection against the water and oxygen that drive hydrolytic and oxidative damage. The megayear numbers, by contrast, are *cold* extrapolations on both sides: Allentoft's 1.5 Myr is at −5 °C and Grass's million-year figure at −18 °C. Both are freezer temperatures, and a shadowed deep-space vault sits far below either. So it is the probe's command of temperature that buys the megayear range, and encapsulation that buys margin against water, oxygen, and handling — not the other way round.

Renewability. The decisive property is the one a self-replicating probe is uniquely positioned to use. A probe that can synthesize and sequence DNA in situ — a capability already implied by the bootstrapping and biological-capability discussions elsewhere in the series — need not treat its archive as a sealed vault to be read once in a million years. It can periodically read, validate, repair, and re-synthesize the archive, replacing degraded strands with fresh copies. Storage becomes a *maintained subsystem* rather than a relic, and its capacity grows with the settlement rather than being fixed at launch. This is the property that lets the physical and logical layers below merge into a self-renewing whole.

The literature behind the renewability claim is thinner than that behind density and durability. Molecular data storage has developed almost entirely as passive cold archiving, with the macromolecules serving as static archives; the reconfigurable operations renewability assumes — erasing, rewriting, repairing, and sorting encoded strands — have only recently been surveyed as a research direction in their own right, across both DNA and synthetic sequence-defined polymers (Ossowski et al. 2026). Renewability is therefore the property this architecture leans on hardest and the one least demonstrated: a design requirement placed on the probe's chemistry, not a capability inherited from the terrestrial state of the art.

4. Coding, Encapsulation, and Redundancy

A megayear archive must survive not just bulk decay but the specific failure modes of DNA — strand breaks, base substitutions, insertions, and deletions. The defenses are layered: error-correcting codes (Reed–Solomon and fountain codes) recover the logical payload from physically damaged strands (Goldman et al. 2013; Erlich & Zielinski 2017; Ceze, Nivala & Strauss 2019); content addressing organizes the archive as a set of Merkle-rooted blocks so that any block is independently verifiable and any corruption is localized; and physical redundancy — many copies, encapsulated and spatially separated — guards against catastrophic loss. The read side of that refresh cycle has an existing discipline behind it. Ancient-DNA work routinely recovers sequence from heavily fragmented, chemically damaged templates, using silica-based and single-stranded extraction tuned for ultrashort fragments, damage-aware library preparation, and bioinformatic pipelines built for short, error-prone reads (Luvanga et al. 2026). That literature also supplies the authentication criteria a self-renewing archive needs before it rewrites anything — fragment-length distribution, nucleotide misincorporation patterns, and contamination controls — which are what distinguish an authentically decayed strand from a corrupted or foreign one. It offers no longevity figure, and nothing in it bears on the half-life and megayear-extrapolation estimates above; its contribution is to the reading and authentication step, not to how long the molecule lasts.

The coding overhead required for megayear integrity is substantial but affordable given DNA's density, and the same Merkle structure that organizes the data for retrieval is, as the next sections show, exactly what the integrity ledger commits to. Which code to use is not a settled question: a standardized benchmark of contemporary and classical DNA-storage codecs, scored against the DNA Data Storage Alliance's consensus criteria, finds that none of the evaluated codecs is optimal across all scored dimensions — information density, error-correction performance across substitutions, insertions, and deletions, throughput, and cost trade off against one another (Anžel et al. 2026). A probe weights that trade space unusually — synthesis throughput and cost matter far less than recovered bits per gram carried and per refresh cycle — so the archive should sit at the high-redundancy end of it, and the codec itself must be recorded in the ledger and encoded self-describingly, or a distant replica will inherit strands it cannot decode.

5. The Mission Ledger: A Logical Integrity Layer

Durability preserves the bits; it does not, by itself, prove they are the *right* bits, authored by a legitimate node and inherited through an authorized chain. That is the job of the **mission ledger**: an append-only, cryptographically authenticated record that each settled probe maintains and that propagates through the lineage. The ledger holds the compact, security-critical commitments — the mission-core hash, the current archive Merkle roots, provenance records, lineage proofs that authenticate a probe as a legitimate descendant, the versions of software and of the interpretive layer, signed updates together with the record of changes that were rejected or quarantined, and delayed consensus checkpoints. Bulk scientific and cultural data stay off-ledger in the Merkle-addressed DNA archives of Section 4; the ledger stores only hashes, provenance, and version history, so verifying that an archive or a replica is what it claims to be reduces to recomputing hashes and checking signatures.

It is worth being precise about the lineage of this idea, because "blockchain" carries baggage that does not fit. The essential mechanism — a hash-linked, append-only, tamper-evident record — was introduced by Haber and Stornetta (1991) for digital timestamping, well before and independent of cryptocurrency, and it rests on cryptographic hash trees (Merkle 1987). A proof-of-work currency blockchain (Nakamoto 2008) is one elaboration of that idea, optimized for a setting — open membership, adversarial economics, low-latency global consensus — that has nothing in common with an interstellar probe lineage. Our ledger keeps the hash-linked, authenticated, append-only core and discards the mining, the economic incentives, and the assumption of real-time agreement. It is, in the abstract's phrase, blockchain-*derived*, but it is closer in spirit to a deep-space digital notary than to a distributed cash system.

6. Consensus under Interstellar Latency

The defining constraint is that real-time global consensus is impossible. Settlements are separated by years to thousands of years of light travel; contact is sparse and occurs through beamed links or physical messenger probes, after the delay-tolerant, store-carry-and-forward fashion developed for an interplanetary internet (Burleigh et al. 2003; Cerf et al. 2007), here scaled to the Galaxy and to millennial delays. No quorum can be assembled on any human or even civilizational timescale.

The ledger therefore abandons the goal of immediate finality and adopts its opposite: **delayed finality, fork tolerance, and auditable divergence.** Settlements that lose contact are *expected* to fork — to extend their ledger histories independently — and this is benign rather than catastrophic, because each fork remains a hash-linked, signed, tamper-evident record. When contact later resumes, the divergent histories can be compared, their provenance audited, and, where appropriate, reconciled into a common history, with disagreements quarantined rather than silently overwritten. High-impact changes — and especially inheritable changes to software or the interpretive layer — can be made to require independent confirmation or quorum agreement across several lineages before they are treated as canonical, a deliberately Byzantine-fault-tolerant posture (Lamport, Shostak & Pease 1982) given that, over megayears, some nodes will certainly malfunction or be compromised. The system's goal is not consensus but *eventual consistency with provenance*: a guarantee not that all lineages agree now, but that whatever they hold is authenticated, attributable, and reconcilable.

7. The DNA-Backed Ledger: Joining the Two Layers

The architecture's central move is to encode the ledger itself in DNA, alongside the archives it commits to. This is not a trivial choice of medium; it is what unifies the two forms of continuity. Because the ledger stores only compact commitments — hashes, signatures, provenance, and version history, not bulk data — it is cheap to synthesize in many redundant copies, and because it is encoded in the same durable, renewable substrate as the archive, a strand recovered after deep time carries *both* a fragment of the archive and the cryptographic commitment that authenticates it. The result is a **self-verifying strand**, and from it a self-verifying archive: any recovered piece can be checked against its own embedded Merkle root and provenance signature without reference to any *surviving institution*. The one external element required is the founding public key, which is itself replicated across archive fragments rather than held by an authority. The property is precise and worth stating narrowly: a co-located Merkle root proves the block has not decayed, and the signature proves it was authored by a holder of the founding key. Neither proves anything against a party able to rewrite the block *and* recompute its commitment. Self-verification of this kind is proof against decay and against forgery by anyone lacking the founding key; it is not proof against a node that has been fully compromised. Durability and authenticity travel together in the same molecule.

This synthesis also closes the renewability loop in a security-preserving way. When the probe periodically re-synthesizes degrading strands (Section 3), it does not merely copy bits; it re-validates each block against the ledger's commitments before rewriting it, and it appends the refresh event to the ledger. Memory maintenance and integrity verification become the same operation. The archive thus repairs and replicates itself under continuous cryptographic check, mirroring at the level of *information* what self-repair and self-replication do at the level of the *probe* — which is the sense in which the title's "self-renewing archive" is meant literally rather than metaphorically.

8. Three Storage Regimes

The renewability property admits a spectrum of operating regimes, and a real probe would likely use all three for different tiers of data.

Passive cold vault. Encapsulated, shielded, cooled DNA, written once and read rarely, relying on Section 3's durability to survive on the order of a million years untouched. This is the safest regime for the most precious and least-changing data — the mission core and the foundational archive — and it requires no working biochemistry between reads.

Actively refreshed archive. The probe periodically reads, validates, repairs, and re-synthesizes the archive, holding integrity indefinitely at the cost of keeping synthesis-and-sequencing machinery alive across deep time. This is a settlement-phase capability, and saying so resolves an ambiguity the three regimes otherwise leave open:

the subsystem budget paper powers the synthesis and sequencing hardware down during cruise, so a probe in transit — 2,800 years to Proxima, some 13,000 on a 20 ly hop — is necessarily in the passive-vault regime whether or not it was designed to be. That is the regime the vault exists for, and the cold, dry, shielded conditions of cruise are exactly the ones the durability argument above requires. The refresh *interval* at settlement is not fixed here; it follows from the survival horizon and the coding scheme's tolerance for strand loss, and it is a far slower clock than any network cadence — refresh is not paced by the lineage-network paper's T_silence of 750–5,000 years. This is the natural regime for the working archive and the ledger, and it is the one that makes "self-renewing" literal; its dependency is the longevity of the read/write apparatus, which couples this subsystem to the probe's self-repair and in-situ manufacturing.

Living archive. The most speculative regime encodes selected information into the genomes of self-reproducing organisms, which copy it as they divide — demonstrated in principle for digital data written into living bacterial populations (Shipman et al. 2017). Such an archive could be extraordinarily robust and self-propagating, but it introduces mutation and selection, which threaten fidelity, and it raises *biological containment* as a first-order concern — the same ethical fork as deliberate life-seeding (directed panspermia) discussed in the bootstrapping work, and one we flag rather than resolve. We regard the living archive as a frontier option, suitable at most for robustly error-corrected, low-priority, deliberately contained payloads, not for the mission core.

9. What the Architecture Secures, and What It Does Not

Honesty about scope is essential, because the architecture is powerful in a narrow way. It secures **syntactic** integrity and **historical accountability**: it makes tampering evident, proves the provenance and version history of archives and updates, authenticates lineage membership, and guarantees that whatever a probe holds is attributable and reconcilable. These are real and non-trivial guarantees, and they are exactly the substrate on which any higher governance must rest.

It does not, by itself, secure several things it must not be mistaken for securing. It cannot guarantee **semantic** integrity — that the meaning of the mission core is preserved as interpreters evolve, which is the job of the payload paper's interpretive layer. It cannot guarantee **value alignment** or correct behavior — an authenticated update can still be a harmful one. It cannot solve **governed amendment** of the mission core — deciding *whether and how* the immutable core may ever be revised is a governance problem the ledger can record and audit but not adjudicate. And it cannot solve **biological containment** for the living-archive regime. The ledger proves that a change happened, by whom, and that the record is intact; it does not prove that the change was wise. We state this plainly because the chief failure mode of integrity systems is to be trusted for more than they provide: an unbroken hash chain is not an unbroken mission.

A further limit follows from who does the logging. Entries recording changes to the cognitive layer are signed by the probe that is making them, so the audited party and the auditor are the same system — the configuration Nagaraj (2026) formalizes as the model-hiding problem for continually self-adapting systems, showing that a tamper-evident Merkle chain over the sequence of updates constrains but never eliminates an adversary's ability to withhold updates that would trigger review. Tamper-evidence proves the recorded history is intact; it does not prove the history is complete, and over megayears the weaker link is the assumption that every self-modification is offered to the chain at all.

Further gaps belong in this accounting, though they sit in the surrounding decisions rather than in the ledger's mechanics. The three-regime structure of Section 8 requires someone to decide what qualifies for the passive vault as mission core and foundational archive versus what is actively refreshed, deprioritized, or never archived at all — and that triage is an act of cultural and scientific curation, not merely of storage engineering. Whose science, whose art, whose history is judged permanent enough to survive the probe is a canon-forming decision the architecture presupposes but never assigns to a decider or a criterion; it is distinct from the question, addressed in the ethics paper, of whose behavioral values are encoded in the immutable core, since curatorial selection concerns content rather than conduct, and it remains as open here as it is unnamed. Separately, the living archive of Section 8 and the renewable synthesis-and-sequencing capability of Section 3 are not only a containment problem in the sense already flagged above: a probe that carries working DNA synthesis and sequencing apparatus, and that can encode data into the genomes of self-reproducing organisms (Shipman et al. 2017), carries a biosecurity and dual-use risk of a categorically different kind than the mechanical self-replication risk the series addresses elsewhere through manufacturing-scope limits. Metal and electronics manufacturing bounded by scope constraints is not pathogen-adjacent or weapons-adjacent in the way that working synthetic-biology apparatus necessarily is, and nothing in this paper's containment caveat for the living archive extends to cover that broader dual-use exposure.

A distinct loss mode sits underneath all of these. An archive can survive physically intact and still become unreadable, because the specification or the competence needed to decode it has decayed — a failure the integrity machinery cannot detect, since the bits verify perfectly against commitments that are themselves intact. This is not bit-rot, and it is not the knowledge-growth paper's L_retrieval, which concerns finding an item rather than decoding one. Digital preservation names the requirement directly: an archived object is interpretable only alongside the *representation information* needed to render it, and that information is itself subject to obsolescence (CCSDS/ISO 14721). Section 4's self-describing-codec requirement is a partial defense against the format half of this; the competence half — the reading system's own interpretive drift — belongs to the payload paper's interpretive layer and the speciation paper's drift clock.

A further gap concerns not what is archived or how it is made, but who eventually reads it. Section 11 names the Rosetta problem of encoding for a recipient with no shared context, but the harder feature of that problem is its one-sidedness: unlike the encounters the governance paper analyzes, in which contact is interactive and meaning can be tested, corrected, and renegotiated across an exchange, a DNA archive recovered after megayears is a message that must be understood correctly on first receipt, by a reader who may be millions of years removed, with no channel back to the sender and no opportunity to observe or repair a misreading. The nearer precedent is not first contact but long-term hazard-marking on Earth: the nuclear-waste semiotics literature that produced the "atomic priesthood" proposal for warning markers meant to remain intelligible ten thousand years hence (Sebeok 1984) confronted exactly this asymmetry, and proposed an institutional relay — a designated custodial body that would re-encode and retransmit the warning across generations, supported by redundant channels of ritual and folklore — concluding that periodic renewal by a custodian, rather than any single ingeniously designed message, was the workable defense. The parallel to the actively refreshed archive of Section 8 is close: both replace the durable message with a maintained one. The mission ledger can authenticate that a recovered record is intact and attributable; unlike its own fork reconciliation for divergent histories, it has no mechanism for reconciling a divergent reading, because by the time one occurs there is no longer anyone on the other end to reconcile with.

10. Integration with the Architecture and the Series

The DNA-backed ledger is the physical and logical home for several mechanisms the other papers assume. The payload paper's integrity ledger and interpretive-layer versioning are realized here; its tiered-retention policy maps onto the three regimes of Section 8, with the mission core in a passive vault and the working archive actively refreshed; and the network governance sketched in the payload paper — provenance, quorum validation, quarantine, fork reconciliation — is built out in the lineage-network paper and runs over precisely this ledger across the delay-tolerant network. The seed archive of the payload paper, with its physics-and-mathematics Rosetta key and its record of origin, is committed into DNA under this scheme, so that a finder recovering the archive recovers its authentication with it. In the series' framing, this paper completes the *memory* subsystem of a probe whose task is to remain itself, keep learning, and carry knowledge forward: durability lets it carry knowledge forward, the ledger lets it remain itself, and renewability lets it do both indefinitely.

11. Open Problems and Future Work

Several problems are named here and left open. The reliability of synthesis and sequencing apparatus over megayears, and its own maintenance, is the central engineering unknown, and it ties memory directly to the self-repair and manufacturing problems of the engineering papers. Cryptographic obsolescence is a second unknown of the same shape: the ledger's hash and signature primitives are assumed to hold over megayears, an assumption no deployed primitive has yet been asked to satisfy — MD5 and SHA-1 were each broken within roughly two decades of wide deployment, and migration away from them took longer than their useful lives; a deep-time ledger needs a governed path for migrating its primitives — re-anchoring old commitments under new ones before the old are broken — and that migration machinery is itself part of the attack surface. The radiation tolerance of encapsulated DNA over megayears, and the energy budget of periodic active refresh, need quantification. The living-archive regime needs both a fidelity model — how much mutation accumulates, and how error correction fares against selection — and a containment framework. The consensus model needs formalization: what "delayed finality" and "eventual reconciliation" mean precisely under millennial latency, and how forks are merged when they conflict. And the deepest problem the ledger deliberately does not solve, governed amendment of the mission core, remains the open governance question the series returns to repeatedly. Finally, the archive must be encoded for decodability by a recipient with no shared context — the Rosetta problem — now compounded by the need to make the *authentication* scheme itself self-describing in DNA.

12. Conclusion

A deep-time interstellar probe must preserve not only its data but its identity, and across megayears and generations of imperfect copies neither ordinary storage nor ordinary trust survives. We have proposed joining two answers into one architecture: synthesized DNA for dense, durable, and — crucially — renewable physical memory, and a hash-linked, append-only mission ledger for cryptographic continuity of identity, provenance, and inheritance, with the ledger encoded into the same DNA it protects so that recovered memory is self-verifying. Designed for sparse, light-delayed contact, the system favors delayed finality, fork tolerance, and eventual reconciliation over impossible real-time consensus, and it spans passive vaults, actively refreshed archives, and speculative living archives. It secures syntactic integrity and historical accountability across deep time while leaving semantic interpretation, value alignment, biological containment, curatorial selection, log completeness, and governed amendment as separate problems built atop it. The result is neither archival storage alone nor blockchain in space, but an integrated, self-renewing memory architecture for autonomous interstellar lineages — a memory that repairs and replicates itself, as the probe does, so that a machine civilization launched from Earth can remain itself, keep learning, and carry what it learns forward across deep time.

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